

A Major Project Report on

**Hybrid Cascade-CNN Architecture in Multi-Level Smart Parking
Environments Integrating RealTime License Plate Recognition for
Automated Access Control and Dynamic Spot Allocation**

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the
academic requirements for the award of the degree of

Bachelor of Technology

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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2025- 2026

DECLARATION

We hereby declare that the project work titled "**Hybrid Cascade-CNN Architecture in Multi-Level Smart Parking Environments Integrating RealTime License Plate Recognition for Automated Access Control and Dynamic Spot Allocation**" is an original work carried out by us in partial fulfillment for the award of Bachelor of Technology in ELECTRONICS AND COMMUNICATION ENGINEERING. This project report has been prepared based on our research and findings and has not been submitted to any other university or institute for the award of any degree or diploma.

We confirm that the work presented in this report is authentic and free from any form of plagiarism. Any references or sources used have been duly acknowledged.

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CERTIFICATE

This is to certify that the Major Project report entitled " Hybrid Cascade-CNN Architecture in Multi-Level Smart Parking Environments Integrating RealTime License Plate Recognition for Automated Access Control and Dynamic Spot Allocation" being submitted by K. Lasya Priya (22H51A0432), J.Anulatha (22H51A04N1), G. Sravya (22H51A0426) in partial fulfillment for the award of Bachelor of Technology in ELECTRONICS AND COMMUNICATION ENGINEERING is a record of Bonafide work carried out under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree.

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ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express our heartfelt gratitude to all the people who helped in making this project a grand success.

We are grateful to **P. Annapurna Bai**, Assistant Professor, Department of Electronics and Communication Engineering, for her valuable technical suggestions and guidance during the execution of this project work.

We express our sincere gratitude to **Dr. J. Seetaram**, Project Coordinator, Electronics and Communication Engineering Department, CMR College of Engineering & Technology, for his continuous guidance, encouragement, and support throughout the completion of our project.

We would like to thank, **Dr. P. Raveendrababu**, Head of the Department of Electronics and Communication Engineering, CMR College of Engineering & Technology, who is the major driving forces to complete our project work successfully.

We are very grateful to **Dr. Ghanta Devadasu**, Dean-Academics, CMR College of Engineering & Technology, for his constant support and motivation in carrying out the project work successfully.

We are highly indebted to **Dr. A Seshu Kumar**, Principal, CMR College of Engineering & Technology, for unwavering support, and encouragement throughout the duration of this project.

We wish to extend our deepest gratitude to the esteemed Director **Major Dr. V A Narayana**, CMR College of Engineering & Technology, for giving permission to carry out this project in a successful and fruitful way.

We would like to thank the **Teaching & Non- teaching** staff of Department of Electronics and Communication Engineering for their co-operation.

We express our sincere thanks to **Shri. Ch. Gopal Reddy**, Secretary& Correspondent, CMR Group of Institutions, and **Shri Ch Abhinav Reddy**, CEO, CMR Group of Institutions for their continuous care and support.

Finally, we extend thanks to our parents who stood behind us at different stages of this Project. We sincerely acknowledge and thank all those who gave support directly or indirectly in completion of this project work.

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ABSTRACT

The challenges associated with managing critical capacity in urban parking infrastructure are currently experienced in the exponential growth of vehicle density and constant limitations associated with available parking space, leading to congestion on the roads, fuel wastefulness, and frustration among drivers. Current smart parking systems use disjointed methods where one considers vehicle detection or license plate recognition separately, and do not provide any architectures that can support simultaneous access control and intelligent allocation of spots in a multi-level parking utility. The study proposed a hybrid cascade-CNN model that combines vehicle detection on the basis of YOLOv8 with a new lightweight convolutional neural network (license plate recognition), together with adaptive spot assigning algorithms in the distribution of parking among multiple facility levels. The given system offers three steps processing pipeline consisting of vehicle detection with a 98.2% accuracy score, license plate localization with a 96.7% precision score, and optical character recognition with a 94.8% character-level score at different illumination levels. In validation in a 500-space multi-level parking unit, it was shown that the average vehicle processing time was 127 milliseconds which then calculated real-time throughput of 28 vehicles per minute at the car entry points. Dynamic allocation algorithm cut the mean parking search time by 43.6 percent as compared to the static assignment mechanism whilst obtaining efficiency of 94.2 percent space utilization. The smart city parking management offers a scalable solution to intelligent parking management in the city.

CHAPTER 1

INTRODUCTION

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1.1. Problem Statement

Urban parking infrastructure faces severe operational challenges driven by rapid vehicle density growth and static space capacity. Studies report that urban drivers spend an average of 17.3 minutes per trip searching for a parking spot, contributing to approximately 32% of urban traffic congestion and generating economic losses exceeding US \$73 billion annually. These inefficiencies stem directly from the architectural limitations of current parking management systems rather than physical space scarcity alone. Existing smart parking deployments treat vehicle detection, license plate recognition, and space allocation as isolated non-interoperable modules, introducing 23–31% operational inefficiencies compared to integrated systems. Static space assignment strategies consistently achieve utilization efficiencies of only 71–78%, while server-based deep learning inference introduces latencies of 200–400 ms, rendering such approaches infeasible for real-time entry-point operations. License plate recognition accuracy further degrades by 12–18% under varying illumination, occlusion, and diverse plate formats. No existing architecture unifies all three functions within a single real-time edge-deployable framework, representing the core gap this work addresses.

1.2. Project Objective

- ❖ To develop an integrated smart parking system combining vehicle detection and license plate recognition.
- ❖ To enable real-time automated access control using accurate number plate identification.
- ❖ To implement a dynamic parking space allocation system based on real-time availability.
- ❖ To reduce parking search time, traffic congestion, and improve overall space utilization.

1.3 Project Scope and Limitations

The proposed hybrid cascade-CNN system is scoped to multi-level urban parking facilities with structured entry/exit control points and pre-registered vehicle databases. The system addresses three integrated functions — vehicle detection, license plate recognition, and dynamic spot allocation — within a unified real-time framework validated on a 500-space, five-level commercial facility over a 30-day continuous deployment period covering 25,847 vehicle entries.

The architecture is designed for edge-deployment on embedded GPU platforms (NVIDIA Jetson Orin NX) with Gigabit Ethernet connectivity to a central PostgreSQL database, supporting facilities of up to 100,000 registered vehicles. Performance was evaluated across three illumination conditions (daytime natural light, nighttime LED, and twilight transition) and four vehicle categories (sedans, SUVs, pickup trucks, and compact cars), with standard, vanity, and temporary regional plate formats included.

Several limitations were identified during field evaluation. The system exhibited reduced recognition accuracy under heavy rainfall and direct sun glare, contributing to the majority of the 48 false rejection incidents recorded. Non-standard and severely damaged license plates accounted for a significant portion of recognition failures, as the current preprocessing pipeline does not incorporate motion-deblurring or adverse-weather augmentation at inference time. The dynamic allocation algorithm, while operating under 15 ms for 1,000+ space facilities, assumes known vehicle dimensions and does not currently support real-time integration with municipal traffic management infrastructure. Additionally, the end-to-end pipeline accuracy of 90.3% — a product of sequential stage-wise error propagation — indicates that detection or localization failures in upstream stages directly limit overall recognition throughput. The system was validated in a single shopping-centre deployment context; generalizability to open-air facilities, underground parking, or facilities with non-standardized lane geometries remains to be established.

CHAPTER 2

LITERATURE SURVEY

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LITERATURE SURVEY

2.1 Introduction

Smart parking systems have gained significant importance due to increasing urban traffic congestion and inefficient space utilization. Various research works have focused on improving parking management using technologies such as IoT, computer vision, deep learning, and optimization algorithms. However, most existing systems address only individual components such as vehicle detection, license plate recognition, or parking allocation, without providing a unified solution.

The rapid growth of urbanization and the increasing number of vehicles have created significant challenges in parking management systems. Finding an available parking space has become a time-consuming and inefficient process, leading to traffic congestion, fuel wastage, and environmental pollution. Traditional parking systems rely on manual monitoring and lack real-time information, making them ineffective in handling modern urban demands.

To address these issues, Smart Parking Systems have been developed by integrating advanced technologies such as the Internet of Things (IoT), Wireless Sensor Networks (WSN), Computer Vision, and Artificial Intelligence (AI). These systems aim to automate parking operations by providing real-time information about parking availability, improving efficiency, and reducing human intervention.

Early smart parking solutions primarily focused on IoT-based sensor networks, where sensors were deployed to detect the occupancy status of parking spaces. Although these systems are effective in identifying whether a parking slot is occupied or vacant, they lack the capability to identify vehicles or perform intelligent decision-making.

With advancements in technology, computer vision and deep learning techniques such as Convolutional Neural Networks (CNN) and YOLO (You Only Look Once) have been widely used for vehicle detection. These methods provide higher accuracy and enable real-time monitoring using camera-based systems. In addition, License Plate Recognition (LPR) systems have been introduced to enable automatic vehicle identification and access control, improving security and management efficiency.

Recent research has also focused on intelligent parking allocation techniques using machine learning and optimization algorithms. These systems analyze real-time data and dynamically assign parking spaces to vehicles, thereby reducing search time and improving space

utilization. However, most existing approaches address only individual components such as detection, recognition, or allocation, rather than providing a complete integrated solution.

Despite these advancements, several challenges still exist. These include high computational requirements, sensitivity to environmental conditions such as lighting and weather, and difficulties in achieving real-time performance in large-scale parking environments. Additionally, many systems lack scalability and fail to integrate all functionalities into a single unified framework.

Therefore, there is a strong need for an integrated smart parking system that combines vehicle detection, license plate recognition, and dynamic parking allocation into a single efficient architecture. This literature survey reviews existing approaches, identifies their limitations, and highlights the research gap addressed by the proposed system.

[1] IoT-Based Parking Monitoring

Rahman and Ahmed (2023) proposed an IoT-enabled smart parking monitoring system based on Wireless Sensor Networks (WSN) utilizing LoRaWAN communication technology. The primary objective of their work was to design a scalable, energy-efficient, and cost-effective solution for real-time parking occupancy detection in large urban environments. The system architecture consists of distributed sensor nodes installed at individual parking slots, which continuously monitor vehicle presence using proximity or magnetic sensors. These sensor nodes transmit occupancy data to a central gateway using LoRaWAN, a long-range, low-power communication protocol particularly suitable for smart city applications.

The methodology focuses on integrating low-power hardware with cloud-based data processing. Each sensor node operates with minimal energy consumption, enabling extended battery life and reducing maintenance requirements. The collected data is transmitted to a centralized server where it is processed and displayed through a user interface, allowing drivers and administrators to monitor parking availability in real time. Experimental evaluation demonstrated an accuracy of approximately 94.7% in detecting occupied and vacant parking spaces. Additionally, the system showed strong scalability, making it suitable for deployment in large infrastructures such as airports, shopping malls, and metropolitan parking facilities.

One of the key strengths of this approach is its energy efficiency and long communication range, which significantly reduces infrastructure and operational costs. The use of LoRaWAN ensures reliable data transmission over large distances without requiring extensive network infrastructure. Furthermore, the system provides continuous monitoring and can be integrated with mobile applications to enhance user convenience.

However, the system has notable limitations. It is restricted to occupancy detection and does not provide any mechanism for vehicle identification, license plate recognition, or automated access control. This limits its applicability in security-sensitive environments. Additionally, the reliance on physical sensors increases installation complexity and cost, and sensor failures due to environmental factors such as dust, water, or mechanical damage can affect system reliability. The system also lacks intelligent decision-making capabilities such as dynamic parking allocation.

This work is highly relevant to the proposed hybrid cascade-CNN architecture, as it highlights the importance of scalable and real-time parking monitoring. However, the proposed system extends beyond occupancy detection by integrating computer vision-based vehicle detection, license plate recognition, and intelligent parking allocation into a unified framework. This eliminates the need for physical sensors while providing enhanced functionality, automation, and security.

[2] Edge-Based CNN Parking Detection

Fernandez et al. (2024) developed an edge-based smart parking detection system using a lightweight convolutional neural network built on the MobileNetV3 architecture. The primary goal of this work was to achieve real-time parking occupancy detection with minimal latency by leveraging edge computing rather than relying on centralized cloud processing. This approach significantly reduces communication delays and improves system responsiveness, making it highly suitable for time-critical applications such as smart parking management.

The system operates by capturing video streams from cameras installed in parking areas and processing them locally on embedded edge devices. The MobileNetV3 model is specifically chosen due to its efficiency, low computational requirements, and suitability for deployment on resource-constrained hardware. The model classifies parking slots as either occupied or vacant by analyzing visual features extracted from the input frames. Experimental results demonstrated an accuracy of 96.3% with an average processing latency of approximately 34 milliseconds, indicating strong real-time performance.

A significant advantage of this system is its independence from cloud infrastructure, which ensures continuous operation even in environments with limited or unstable internet connectivity. Edge-based processing also enhances data privacy, as sensitive video data does not need to be transmitted to remote servers. Additionally, reduced bandwidth usage makes the system more cost-effective for large-scale deployments.

Despite these strengths, the system has certain limitations. It focuses solely on parking

occupancy detection and does not incorporate vehicle identification or license plate recognition capabilities. As a result, it cannot be used for automated access control or security monitoring. Furthermore, the system does not include any intelligent parking allocation mechanism, limiting its ability to optimize parking space utilization. Performance may also degrade under challenging conditions such as poor lighting, shadows, or occlusions, which are common in real-world parking environments.

Another limitation is the lack of integration with higher-level decision-making modules. While the system efficiently detects occupancy, it does not provide guidance or allocation strategies for incoming vehicles. This reduces its effectiveness as a complete smart parking solution.

This work is directly relevant to the proposed system, as it demonstrates the effectiveness of edge computing for real-time applications. The proposed hybrid cascade-CNN architecture builds upon this concept by integrating additional modules such as license plate recognition and dynamic parking allocation. By combining detection, identification, and decision-making within a unified framework, the proposed system addresses the limitations of standalone detection models and provides a comprehensive smart parking solution.

[3] YOLOv8 Optimization for Parking-Specific Vehicle Detection

Nguyen and Tran (2024) proposed an optimized vehicle detection system based on the YOLOv8 architecture, specifically tailored for parking environments. The primary objective of this research was to improve detection accuracy and robustness in complex real-world parking scenarios, including dense vehicle arrangements, partial occlusions, and varying illumination conditions. YOLOv8, being a one-stage object detection model, processes the entire image in a single forward pass, enabling high-speed inference suitable for real-time applications such as smart parking systems.

The methodology involves training the YOLOv8 model on a customized dataset containing parking-specific vehicle images captured under diverse environmental conditions such as daytime, nighttime, and shadowed areas. The model utilizes multi-scale feature extraction and feature pyramid networks to detect vehicles of different sizes and orientations. Additionally, data augmentation techniques such as flipping, scaling, and brightness adjustments are applied to improve model generalization. The optimized YOLOv8 model demonstrated a detection accuracy of 97.9% with low inference latency, making it highly effective for deployment at parking entry and exit points.

One of the major strengths of this approach is its ability to perform real-time detection with high precision, even in crowded and cluttered environments. The model's architecture allows

it to detect multiple vehicles simultaneously, which is essential for multi-level parking systems. Furthermore, its adaptability to different environmental conditions enhances its practical applicability.

However, the system has certain limitations. It focuses solely on vehicle detection and does not incorporate license plate recognition or parking allocation mechanisms. As a result, it cannot function as a complete smart parking solution. Additionally, the model's performance depends on the quality and diversity of the training dataset, and retraining may be required for different deployment environments.

This work is directly relevant to the proposed hybrid cascade-CNN architecture, as YOLOv8 is used as the primary detection module. The proposed system extends this work by integrating detection with license plate recognition and dynamic allocation, thereby providing a complete end-to-end solution.

[4] Spatial Transformer Networks for License Plate Recognition

Silva and Costa (2024) introduced a license plate recognition system based on Spatial Transformer Networks (STN), aiming to address the challenges caused by geometric distortions in captured license plate images. The primary objective of this work was to improve recognition accuracy by automatically correcting distortions such as rotation, skew, and perspective variations, which commonly occur due to different camera angles and vehicle positions.

The methodology integrates an STN module within a deep neural network architecture. The spatial transformer learns transformation parameters that align the input image into a normalized format before it is passed to the recognition stage. This eliminates the need for manual preprocessing steps such as cropping or rotation correction. After normalization, a convolutional neural network extracts features from the aligned plate image, and an OCR module is used to recognize the characters. Experimental results show that the system significantly improves recognition accuracy, especially for distorted or angled license plates. A key strength of this approach is its ability to handle real-world challenges related to image capture conditions. By incorporating geometric transformation directly into the network, the system enhances robustness and reduces dependency on external preprocessing techniques. This makes the model more flexible and adaptable to different environments.

However, the system also has limitations. The inclusion of spatial transformation modules increases computational complexity, making it less suitable for real-time edge deployment where resources are limited. Additionally, the system focuses only on recognition and does not

include vehicle detection or parking allocation modules. Performance may also degrade in cases of severe occlusion or damaged license plates.

This work is relevant to the proposed system as it highlights the importance of plate normalization for accurate recognition. The hybrid cascade-CNN architecture incorporates similar concepts but uses more lightweight techniques to maintain real-time performance.

[5] Transformer-Based Sequence Modeling for License Plate Recognition

Zhou and Wang (2024) proposed a transformer-based optical character recognition system for license plate recognition, focusing on improving sequence modeling using attention mechanisms. The main objective of this research was to overcome the limitations of traditional CNN-based OCR systems by capturing contextual relationships between characters in a license plate sequence.

The methodology involves extracting the license plate region and feeding it into a transformer-based model. Unlike conventional approaches that treat character recognition as independent classification tasks, this model processes the entire sequence simultaneously. The attention mechanism allows the model to focus on relevant parts of the image while predicting each character, thereby improving recognition accuracy. The system demonstrates high performance across different plate formats, including variations in font, spacing, and alignment.

One of the major strengths of this approach is its ability to model long-range dependencies between characters, which enhances recognition accuracy in complex scenarios. The transformer architecture also provides flexibility in handling different sequence lengths and formats. Additionally, the model shows strong performance in noisy and cluttered environments.

However, the system has several limitations. Transformer models are computationally intensive and require large datasets for training, making them less suitable for deployment on edge devices. The inference time is also higher compared to lightweight CNN-based models, which can affect real-time performance. Furthermore, the system does not include vehicle detection or parking allocation modules, limiting its integration into a complete smart parking system.

This work is relevant to the proposed system as it provides insights into advanced OCR techniques. However, the proposed hybrid cascade-CNN architecture adopts a more efficient CNN-BiLSTM approach to balance accuracy and real-time processing requirements.

[6] Reinforcement Learning-Based Dynamic Parking Allocation

Chen and Zhang (2024) proposed a dynamic parking allocation system based on deep

reinforcement learning, aimed at optimizing parking efficiency in urban environments. The primary objective of this work was to develop an intelligent system capable of adapting to real-time conditions and making optimal allocation decisions based on continuous learning.

The methodology models the parking allocation problem as a sequential decision-making process. The system observes the current state of the parking environment, including slot availability and vehicle information, and selects an action corresponding to assigning a parking slot. A reward function is used to evaluate the effectiveness of each decision, encouraging the system to minimize search time and maximize space utilization. Over time, the model learns optimal strategies through repeated interactions with the environment. Experimental results show significant improvements in parking efficiency and reduced search time compared to traditional methods.

One of the key strengths of this approach is its adaptability. The system continuously learns from changing conditions and improves its performance over time. It is particularly effective in dynamic environments where parking demand fluctuates. Additionally, reinforcement learning allows the system to handle complex optimization problems.

However, the system has limitations. It requires extensive training data and high computational resources, making it difficult to deploy in real-time edge environments. The training process can be time-consuming, and performance may vary in unseen scenarios. Furthermore, the system does not integrate detection or recognition modules, limiting its applicability as a standalone solution.

This work is relevant to the proposed system as it highlights the importance of intelligent allocation. The hybrid cascade-CNN system incorporates a computationally efficient allocation strategy inspired by such approaches while maintaining real-time performance.

[7] Graph-Based Optimization Algorithms for Parking Management

Krishnan and Gupta (2023) proposed a graph-based optimization approach for efficient management of multi-level parking facilities. The primary objective of this work was to develop a mathematical framework that optimizes parking space allocation by minimizing travel distance and improving overall utilization. The system models the parking environment as a graph structure, where nodes represent parking slots and edges represent possible paths or transitions between them. This representation allows the application of well-established optimization algorithms to determine the best allocation strategy.

The methodology is based on minimum-cost flow algorithms, where each vehicle is assigned to a parking slot in such a way that the overall cost, typically defined in terms of distance or

time, is minimized. The system considers constraints such as slot availability, vehicle type compatibility, and spatial layout of the parking facility. Simulation results indicate that the graph-based approach significantly improves efficiency compared to traditional random or static allocation methods, particularly in structured environments like multi-level parking systems.

One of the key strengths of this approach is its strong theoretical foundation, which guarantees optimal or near-optimal solutions under defined constraints. The method is computationally efficient for moderately sized systems and can be extended to handle multiple objectives. However, the approach relies heavily on predefined parameters and static data, which limits its adaptability to real-time dynamic conditions. It does not learn from changing patterns or user behavior, making it less effective in highly dynamic environments.

Additionally, the system does not incorporate vehicle detection or license plate recognition, which are essential components for automation. It assumes that all necessary input data is already available, which may not be practical in real-world scenarios.

This work is relevant to the proposed system as it provides a foundation for optimization-based allocation strategies. The hybrid cascade-CNN architecture improves upon this by integrating real-time data from detection and recognition modules, enabling dynamic and adaptive allocation.

[8] CNN-YOLOv4 Based Real-Time Vehicle Detection

Sailaja et al. (2024) presented a real-time vehicle detection system combining Convolutional Neural Networks with the YOLOv4 object detection framework. The objective of this research was to enhance detection accuracy and speed in dynamic environments such as traffic monitoring and parking systems. The integration of CNN and YOLOv4 enables efficient feature extraction and fast inference, making it suitable for real-time applications.

The methodology involves capturing video frames from surveillance cameras and processing them through a CNN-based backbone network for feature extraction. These features are then passed to the YOLOv4 detection head, which predicts bounding boxes and class probabilities for detected vehicles. The system is trained on diverse datasets to improve robustness under varying environmental conditions. Experimental results demonstrate that the system achieves high detection accuracy with real-time performance.

A major strength of this approach is its ability to detect multiple vehicles simultaneously with high precision. The combination of CNN and YOLOv4 provides a balance between accuracy and speed. However, the system requires significant computational resources, which may limit

its deployment on low-power devices.

Another limitation is that the system focuses only on detection and does not include modules for license plate recognition or parking allocation. This restricts its functionality in a complete smart parking system.

This work is relevant to the proposed system as it validates the effectiveness of CNN-YOLO architectures for vehicle detection. The proposed hybrid cascade-CNN system extends this approach by using a more advanced YOLOv8 model and integrating additional functionalities.

[9] Deep Learning-Based Vehicle Monitoring System

Srinivasula Reddy et al. (2024) proposed a deep learning-based system for vehicle monitoring and identification in real-time scenarios. The primary objective of this work was to leverage convolutional neural networks to improve vehicle detection and tracking accuracy in dynamic environments.

The methodology involves using a deep CNN architecture to process video input and extract features for vehicle detection. The system is capable of identifying vehicles and tracking their movement across frames, enabling continuous monitoring. The model is trained on large datasets to improve generalization and robustness. Experimental results indicate improved detection accuracy compared to traditional methods.

One of the key strengths of this approach is its ability to learn complex visual features, enabling accurate detection even in challenging conditions. The system is also adaptable to different environments with proper training.

However, the system requires large datasets and high computational resources, which can limit real-time deployment on edge devices. Additionally, it does not include license plate recognition or parking allocation modules, reducing its applicability in a complete smart parking system.

This work is relevant to the proposed system as it highlights the importance of deep learning in vehicle detection. The hybrid cascade-CNN architecture builds upon this by integrating detection with recognition and allocation.

[10] Attention-Enhanced YOLO for Vehicle Detection

Anderson and Brown (2023) proposed an attention-enhanced YOLO-based vehicle detection model aimed at improving performance in congested parking environments. The objective of this research was to enhance detection accuracy by incorporating attention mechanisms that allow the model to focus on relevant features within an image.

The methodology involves integrating attention modules into the YOLO architecture, enabling

the model to prioritize important regions and suppress irrelevant background information. This improves detection performance in crowded and occluded scenarios. Experimental results show enhanced accuracy compared to standard YOLO models.

A major strength of this approach is its ability to handle complex environments with high vehicle density. The attention mechanism improves feature representation and detection reliability. However, the increased model complexity leads to higher computational requirements, which may affect real-time performance on low-power devices.

Additionally, the system focuses only on detection and does not include recognition or allocation modules. This limits its use in a complete smart parking system.

This work is relevant to the proposed system as it highlights improvements in detection accuracy. The hybrid cascade-CNN system balances such enhancements with computational efficiency for real-time deployment.

[11] Deep Learning Approaches for Automated Vehicle Detection

Chen et al. (2024) conducted a comprehensive study on deep learning approaches for automated vehicle detection, focusing on performance analysis and deployment considerations. The objective was to evaluate different models and identify their strengths and limitations in real-world applications.

The study highlights that CNN-based models can effectively extract hierarchical features, enabling accurate detection across diverse environments. It also discusses optimization techniques such as pruning and quantization to improve efficiency. The analysis shows that deep learning models outperform traditional methods in terms of accuracy.

However, the study also identifies challenges such as high computational requirements and dependency on large datasets. It does not propose a complete integrated system.

This work is relevant as it provides theoretical support for using deep learning in the proposed system.

[12] Augmented Reality Navigation for Parking Guidance

Dubey and Sharma (2023) proposed an augmented reality-based parking guidance system designed to assist drivers in navigating parking facilities. The objective was to improve user experience by providing intuitive visual guidance.

The system overlays navigation instructions onto real-world views using mobile devices. It uses indoor positioning technologies to determine user location and guide them to available parking spaces. Results show reduced search time and improved usability.

However, the system depends on accurate positioning and user interaction. It does not include

detection or recognition modules.

This work is relevant for enhancing user interaction in future extensions of the proposed system.

[13] Field Deployment Challenges in Automated License Plate Recognition

Kumar and Singh (2024) presented a comprehensive study on the challenges involved in deploying Automated License Plate Recognition (ALPR) systems in real-world environments. The primary objective of this work was to analyze the practical issues that arise when transitioning ALPR systems from controlled laboratory conditions to real-world applications such as parking facilities, highways, and urban surveillance systems. The study emphasizes that although modern deep learning models achieve high accuracy in ideal conditions, their performance often degrades significantly in real deployment scenarios due to environmental and operational constraints.

The methodology involves evaluating ALPR systems across multiple real-world conditions, including varying illumination levels, motion blur caused by moving vehicles, occlusions due to other vehicles or objects, and inconsistencies in license plate formats. The authors highlight that lighting variations, such as nighttime conditions or direct sunlight, can severely affect image quality and reduce recognition accuracy. Additionally, factors such as dirty, damaged, or non-standard license plates introduce further challenges in character recognition. The study also examines the impact of hardware-related factors such as camera resolution, positioning, and frame rate on system performance.

One of the major strengths of this work is its focus on practical deployment considerations rather than theoretical performance. The authors provide valuable insights into system design, including recommendations for optimal camera placement, use of high-resolution imaging sensors, and the importance of preprocessing techniques such as contrast enhancement and noise reduction. These insights are critical for improving system robustness in real-world applications.

However, the study has certain limitations. It primarily focuses on identifying challenges and does not propose a complete integrated solution. It lacks implementation details for combining detection, recognition, and decision-making modules into a unified framework. Additionally, it does not address real-time processing constraints for edge deployment.

This work is highly relevant to the proposed hybrid cascade-CNN system, as it highlights the exact real-world challenges that the proposed architecture aims to overcome through robust detection, normalization, and recognition modules designed for varying environmental

conditions.

[14] Illumination-Invariant License Plate Recognition

Hassan and Ali (2024) proposed an illumination-invariant license plate recognition system aimed at improving recognition accuracy under varying lighting conditions. The primary objective of this research was to address one of the major limitations of conventional ALPR systems, which is their sensitivity to lighting variations such as low light, shadows, glare, and uneven illumination.

The methodology involves applying adaptive preprocessing techniques to enhance image quality before performing character recognition. These techniques include contrast enhancement, histogram equalization, normalization, and noise reduction. By improving the visibility of license plate characters, the system enables more accurate feature extraction. The enhanced images are then processed using a deep learning-based OCR model for character recognition. Experimental results demonstrate that the system achieves improved accuracy in both low-light and high-glare conditions compared to traditional methods.

A key strength of this approach is its ability to handle challenging lighting conditions effectively, making it suitable for real-world deployment where lighting is unpredictable. The preprocessing techniques are relatively simple to implement and can be integrated with existing recognition systems. Additionally, the method improves robustness without requiring major changes to the recognition model.

However, the system has certain limitations. The additional preprocessing steps increase computational overhead, which may impact real-time performance. The system also focuses only on license plate recognition and does not include vehicle detection or parking allocation modules. Furthermore, performance may still degrade under extreme lighting conditions or severe image distortions.

This work is relevant to the proposed system as it highlights the importance of preprocessing for improving recognition accuracy. The hybrid cascade-CNN architecture can incorporate similar techniques to enhance robustness while maintaining efficient real-time processing.

[15] Field Deployment Challenges in ALPR Systems

Kumar and Singh (2024) further investigated the deployment challenges associated with ALPR systems, focusing on real-world operational constraints and system reliability. The objective of this work was to evaluate how environmental factors, hardware limitations, and data variability affect the performance of ALPR systems in large-scale implementations.

The methodology includes analyzing system performance under different conditions such as

varying camera angles, motion blur, weather effects, and diverse license plate formats. The study highlights that improper camera placement and low-resolution imaging can significantly reduce detection and recognition accuracy. It also discusses the importance of maintaining consistent data quality and using robust preprocessing techniques to mitigate environmental effects.

One of the strengths of this work is its emphasis on practical system design considerations. The authors provide guidelines for improving performance, including optimizing camera placement, using high-quality sensors, and incorporating preprocessing techniques to enhance image quality. These recommendations are essential for achieving reliable performance in real-world applications.

However, the study does not propose a complete system architecture and mainly focuses on identifying issues and suggesting improvements. It also does not address integration with detection or allocation modules, which are necessary for a complete smart parking system.

This work is relevant to the proposed system as it reinforces the need for robust and reliable design. The hybrid cascade-CNN architecture addresses these challenges by integrating optimized detection, localization, and recognition modules for real-time performance.

[16] User Preference Learning for Parking Allocation

Lee and Park (2024) proposed a machine learning-based parking allocation system that incorporates user preference learning to improve efficiency and user satisfaction. The primary objective of this work was to develop a system capable of predicting user preferences based on historical data and using this information to assign optimal parking slots.

The methodology involves collecting user-related data such as preferred parking locations, time of arrival, frequency of visits, and past behavior patterns. Machine learning algorithms are then used to analyze this data and generate predictive models that determine the most suitable parking slot for each user. The system dynamically assigns parking spaces by combining real-time availability with predicted preferences. Experimental results indicate that this approach reduces search time and enhances overall user experience.

A major strength of this system is its ability to provide personalized services. By learning user behavior over time, the system adapts to individual preferences and improves efficiency. Additionally, it enhances space utilization by distributing vehicles more effectively across available slots.

However, the system has several limitations. It requires a large amount of user data for accurate predictions, which may not always be available. Privacy and data security concerns also arise

due to the storage and processing of user information. Furthermore, the system does not include vehicle detection or license plate recognition modules, limiting its integration into a fully automated parking system.

This work is relevant to the proposed system as it demonstrates the importance of intelligent and adaptive allocation strategies. The hybrid cascade-CNN system can incorporate such techniques in future enhancements while maintaining real-time automation.

[17] Edge Computing Architectures for Real-Time Intelligent Transportation

Li and Chen (2023) proposed an edge computing-based architecture for real-time intelligent transportation systems, with a focus on reducing latency and improving system responsiveness. The primary objective of this work was to shift computational workloads from centralized cloud servers to edge devices located closer to the data source, thereby enabling faster processing and real-time decision-making. This approach is particularly relevant for applications such as smart parking systems, where immediate responses are required for tasks like vehicle detection and access control.

The methodology involves deploying edge devices equipped with computational capabilities near data acquisition points such as cameras and sensors. These devices process data locally, reducing the need for continuous data transmission to remote servers. The system uses optimized deep learning models that can run efficiently on edge hardware while maintaining acceptable accuracy levels. Experimental results demonstrate that edge-based processing significantly reduces latency and improves system performance compared to cloud-based approaches.

One of the major strengths of this system is its ability to provide real-time responses, which is critical for dynamic environments like parking facilities. By minimizing data transmission, the system also reduces network bandwidth usage and enhances data privacy. Additionally, edge computing improves system reliability by enabling operation even during network disruptions. However, the system has certain limitations. Edge devices typically have limited computational resources compared to cloud servers, which restricts the complexity of models that can be deployed. This creates a trade-off between accuracy and efficiency. Furthermore, managing multiple edge devices can increase system complexity.

This work is highly relevant to the proposed system, as the hybrid cascade-CNN architecture is designed for edge deployment. It adopts similar principles to achieve low-latency, real-time processing for vehicle detection and license plate recognition.

[18] Operational Efficiency of Integrated Parking Systems

Martinez and Rodriguez (2023) analyzed the operational efficiency of integrated parking management systems compared to fragmented systems. The primary objective of this study was to evaluate how combining multiple functionalities—such as vehicle detection, identification, and allocation—into a single system affects overall performance and efficiency. The methodology involves comparing integrated systems with traditional fragmented approaches where different modules operate independently. The study evaluates metrics such as processing time, resource utilization, and system coordination. Results indicate that integrated systems significantly reduce operational inefficiencies by eliminating redundant processes and improving communication between components.

One of the key strengths of this work is its emphasis on system-level optimization. The study demonstrates that integration leads to better coordination, reduced latency, and improved resource utilization. It also highlights the importance of designing unified architectures for complex applications like smart parking.

However, the study is primarily analytical and does not propose a specific implementation. It lacks detailed technical solutions for achieving integration in real-world systems. Additionally, it does not address challenges such as scalability and computational constraints.

This work is directly relevant to the proposed system, as the hybrid cascade-CNN architecture is designed as an integrated solution that combines detection, recognition, and allocation into a single framework, addressing the inefficiencies of fragmented systems.

[19] Multi-National License Plate Recognition Systems

Park and Kim (2023) proposed a multilingual and multi-national license plate recognition system using language-agnostic neural architectures. The objective of this research was to develop a recognition system capable of handling diverse license plate formats from different regions without requiring separate models for each format.

The methodology involves training a deep learning model on a dataset containing license plates from multiple countries with varying formats, fonts, and character sets. The system uses a unified architecture that learns general features applicable across different plate styles. Experimental results show that the model achieves high recognition accuracy across multiple regions, demonstrating strong generalization capability.

A major strength of this approach is its flexibility and scalability. The system can be deployed in international environments without requiring region-specific customization. It also reduces maintenance efforts by eliminating the need for multiple models.

However, the system requires large and diverse datasets for training, which can be challenging

to obtain. Additionally, the model may struggle with extremely rare or unconventional plate formats. The system also focuses only on recognition and does not include detection or parking allocation modules.

This work is relevant to the proposed system as it highlights the importance of handling diverse plate formats. The hybrid cascade-CNN system can incorporate similar techniques to support multilingual and region-specific recognition.

[20] Space Utilization Optimization in Multi-Level Parking

Patel and Sharma (2024) proposed an optimization-based approach for improving space utilization in multi-level parking structures. The primary objective of this work was to maximize the efficiency of parking space usage by minimizing unused slots and reducing congestion within parking facilities.

The methodology involves analyzing real-time occupancy data and applying optimization techniques to allocate parking spaces effectively. The system considers factors such as vehicle size, parking duration, and slot availability to determine optimal assignments. Simulation and field validation results indicate significant improvements in space utilization and reduced vehicle search time.

One of the key strengths of this approach is its focus on practical efficiency improvements. By optimizing space usage, the system enhances overall performance and user experience. It is particularly effective in structured environments such as multi-level parking facilities.

However, the system has certain limitations. It relies on accurate input data and does not include mechanisms for vehicle detection or identification. The approach also assumes that all necessary information is available in advance, which may not be realistic in dynamic environments.

This work is relevant to the proposed system as it highlights the importance of efficient space utilization. The hybrid cascade-CNN architecture incorporates dynamic allocation strategies to achieve similar objectives while integrating real-time detection and recognition.

[21] Economic Impact of Parking Search Behavior

Thompson and Williams (2024) conducted a detailed study on the economic impact of parking search behavior in metropolitan areas. The primary objective of this research was to analyze how inefficient parking systems contribute to traffic congestion, fuel consumption, and economic losses. The study highlights that a significant portion of urban traffic is caused by drivers searching for parking spaces, leading to wasted time and increased operational costs.

The methodology involves analyzing traffic data, parking patterns, and fuel consumption

statistics across multiple urban regions. The study quantifies the average time spent searching for parking and its contribution to congestion levels. Results indicate that inefficient parking systems lead to substantial economic losses, including increased fuel costs, reduced productivity, and environmental pollution.

One of the major strengths of this work is its macro-level analysis, which provides a broader understanding of the impact of parking inefficiencies. It highlights the importance of developing intelligent parking systems to address these issues. The study also emphasizes the role of automation and real-time information in reducing search time and improving urban mobility.

However, the study does not propose a specific technical solution or system architecture. It focuses primarily on analysis and does not address implementation challenges. Additionally, it does not integrate detection, recognition, or allocation mechanisms.

This work is highly relevant to the proposed system as it justifies the need for intelligent parking solutions. The hybrid cascade-CNN architecture directly addresses these issues by reducing search time and improving efficiency through real-time detection and dynamic allocation.

[22] Integration Challenges in Automated Parking Systems

Wang and Liu (2023) analyzed the integration challenges associated with end-to-end automated parking systems. The primary objective of this research was to identify the difficulties in combining multiple components such as detection, recognition, and allocation into a unified framework.

The methodology involves studying existing parking systems and evaluating the challenges faced during integration. These challenges include data synchronization, communication delays, compatibility between modules, and scalability issues. The study highlights that many systems fail to achieve full automation due to lack of proper integration.

A key strength of this work is its focus on system-level challenges. It provides valuable insights into the importance of designing unified architectures rather than fragmented solutions. The study emphasizes that integration is critical for achieving real-time performance and system efficiency.

However, the work does not propose a concrete implementation or solution. It mainly focuses on identifying problems rather than solving them. Additionally, it does not address specific techniques for improving integration.

This work is directly relevant to the proposed system, as the hybrid cascade-CNN architecture

is designed to overcome these challenges by integrating detection, recognition, and allocation into a single unified pipeline.

[23] Motion Deblurring for License Plate Recognition

Yamamoto and Suzuki (2023) proposed a motion deblurring technique for improving license plate recognition in moving vehicles. The primary objective of this research was to address the issue of motion blur, which occurs when vehicles move at high speeds, resulting in blurred images that reduce recognition accuracy.

The methodology involves using deep learning-based deblurring networks to restore image clarity before performing recognition. The system processes blurred images and reconstructs sharper versions, enabling better feature extraction and character recognition. Experimental results show that the deblurring approach significantly improves recognition accuracy in high-speed scenarios.

One of the major strengths of this work is its ability to handle dynamic conditions, making it suitable for real-world applications such as highway monitoring and parking entry points. The approach enhances the robustness of license plate recognition systems.

However, the system introduces additional computational overhead, which may affect real-time performance. It also focuses only on preprocessing and recognition, without integrating detection or allocation modules.

This work is relevant to the proposed system as it highlights the importance of handling motion-related distortions. The hybrid cascade-CNN architecture can incorporate such techniques to improve performance under dynamic conditions.

[24] Latency Optimization in Cloud-Based ALPR Systems

Zhang et al. (2024) proposed techniques for optimizing latency in cloud-based license plate recognition systems. The primary objective of this research was to reduce processing delays and improve system responsiveness in cloud-based architectures.

The methodology involves analyzing data transmission delays, server processing time, and system architecture. Optimization techniques such as efficient data compression, parallel processing, and load balancing are applied to reduce latency. Experimental results show improved response time and system efficiency.

One of the key strengths of this approach is its focus on performance optimization in large-scale systems. It provides practical solutions for improving cloud-based processing.

However, the system relies heavily on internet connectivity and cloud infrastructure, which may introduce delays in real-time applications. It also does not include detection or allocation

modules.

This work is relevant to the proposed system as it highlights latency challenges. The hybrid cascade-CNN architecture addresses these issues by performing processing at the edge, reducing dependency on cloud systems.

Table 1 summarizes comparative analysis of existing approaches highlighting capability gaps addressed by the proposed architecture.

TABLE I: COMPARATIVE ANALYSIS OF SMART PARKING SYSTEMS

Reference	Methodology	Detection Accuracy	Recognition Accuracy	Processing Time	Allocation	Limitation
Rahman & Ahmed (2023)	IoT sensor network	94.7%	N/A	180 ms	Static	No identification
Fernandez et al. (2024)	MobileNetV3 vision	96.3%	N/A	34 ms	None	Detection only
Nguyen & Tran (2024)	YOLOv8 detection	97.9%	N/A	28 ms	None	No recognition
Silva & Costa (2024)	STN + attention	N/A	95.4%	156 ms	None	Recognition only
Zhou & Wang (2024)	Transformer OCR	N/A	96.8%	89 ms	None	No detection
Chen & Zhang (2024)	RL allocation	N/A	N/A	N/A	Dynamic	No vision
Krishnan & Gupta (2023)	Graph optimization	N/A	N/A	N/A	Dynamic	Assumes known vehicles
Proposed	Cascade CNN	98.2%	94.8%	127 ms	Dynamic	Integrated

The comparative analysis revealed that existing systems addressed individual components—detection, recognition, or allocation—without end-to-end integration combining all functions within a unified real-time architecture. The proposed hybrid cascade-CNN architecture was

designed to address this integration gap, with each architectural decision — including the selection of BiLSTM-CTC over transformer-based OCR — driven by the dual constraints of real-time throughput and edge-deployment viability rather than benchmark accuracy alone.

2.2 Problem Identification

From the detailed study of existing smart parking systems, vehicle detection models, and license plate recognition techniques, several critical limitations have been identified which highlight the need for a more efficient and integrated solution.

1. Lack of Integration Between Modules: Most existing systems are designed to perform only a single function such as vehicle detection, license plate recognition, or parking allocation independently. There is no proper integration between these modules to form a unified pipeline. As a result, data flow between components becomes inefficient, and additional processing time is required to coordinate between different subsystems. This fragmented architecture reduces system reliability and increases latency, making it unsuitable for real-time applications such as automated parking management. A fully integrated system is required to ensure seamless interaction between detection, recognition, and allocation processes.

2. High Computational Complexity: Many advanced approaches, especially those based on deep learning models like transformers and attention-based networks, provide high accuracy but at the cost of increased computational requirements. These models require powerful GPUs and large memory, which are not feasible for deployment in edge devices or embedded systems commonly used in parking infrastructures. This limits their practical applicability. Moreover, high computational demand leads to increased power consumption and operational cost, making such systems inefficient for large-scale real-time deployment.

3. Sensitivity to Environmental Conditions: Real-world environments introduce several unpredictable factors such as varying lighting conditions, shadows, reflections, rain, fog, and occlusions. These environmental variations significantly affect the performance of detection and recognition systems. For example, low-light conditions reduce visibility, while shadows and reflections can create false detections. Many existing models are trained on controlled datasets and fail to generalize well under such conditions, resulting in reduced accuracy and reliability in practical scenarios.

4. Inaccurate License Plate Recognition: License plate recognition systems often suffer from inaccuracies due to geometric distortions, motion blur, and low-resolution images. Variations in camera angles and vehicle movement cause skewed or blurred images, making character extraction difficult. Although some systems use preprocessing techniques such as image

enhancement or spatial transformation, these methods increase processing time and computational overhead. Additionally, damaged or non-standard plates further complicate recognition, reducing overall system performance.

5. Inefficient Parking Allocation Methods: Traditional parking systems rely on static or rule-based allocation methods that do not adapt to real-time changes in parking availability or traffic conditions. These methods fail to optimize space utilization and often result in congestion and increased search time for drivers. While some advanced approaches use optimization or machine learning techniques, they are often complex and require significant computational resources, making them impractical for real-time applications.

6. Dependency on Additional Hardware: Several existing smart parking systems depend on external hardware such as IoT sensors, RFID systems, or cloud-based infrastructure. While these technologies improve accuracy, they significantly increase system cost, installation complexity, and maintenance requirements. Additionally, reliance on network connectivity makes the system vulnerable to failures due to connectivity issues, reducing reliability and scalability in large parking environments.

7. Lack of Real-Time Processing Capability: Many current systems are not optimized for real-time performance. High processing time due to complex algorithms and data transfer delays leads to latency in detection, recognition, and decision-making. In high-traffic parking areas, even small delays can result in congestion and poor user experience, making real-time capability a critical requirement.

8. Trade-off Between Accuracy and Efficiency: A major challenge in existing systems is balancing accuracy, speed, and computational efficiency. High-accuracy models generally require more computational power and processing time, while lightweight models sacrifice accuracy for faster performance. This trade-off makes it difficult to design a system that performs efficiently in real-time while maintaining high accuracy, which remains an open problem in smart parking system design.

In addition to the above challenges, inefficient parking systems also contribute significantly to urban traffic congestion and environmental pollution. Vehicles searching for parking spaces increase fuel consumption and travel time, leading to economic losses and user frustration. In densely populated areas such as shopping malls, airports, and city centers, the lack of an intelligent parking system results in poor space utilization and overcrowding. These issues highlight the necessity of developing a smart and automated parking solution that can operate efficiently in real-time while minimizing human intervention.

Table 2: Comparison Table of All Research Papers

Ref No	Method Used	Key Features	Advantages	Limitations
[1]	IoT-Based System	Sensor-based detection	Scalable, low power	Requires hardware, high cost
[2]	EdgeCNN (MobileNetV3)	Real-time edge processing	Low latency	No license plate recognition
[3]	YOLOv8 Detection	Deep learning detection	High accuracy	Only detection
[4]	STN-based LPR	Image normalization	Handles distortion	High computational cost
[5]	Transformer OCR	Sequence modeling	High recognition accuracy	Complex and slow
[6]	Reinforcement Learning	Dynamic allocation	Adaptive system	High complexity
[7]	Graph-Based Allocation	Optimization-based method	Efficient allocation	Not real-time adaptable
[8]	CNN + YOLOv4	Hybrid detection	Good performance	High resource usage
[9]	Deep CNN Monitoring	Detection + tracking	Robust detection	Needs large dataset
[10]	Attention YOLO	Feature enhancement	Better accuracy in crowd	Increased computation
[11]	Deep Learning Analysis	Model comparison	Strong theoretical base	No implementation
[12]	AR-Based System	User guidance	Improves user experience	Depends on user interaction
[13]	ALPR Challenges Study	Real-world analysis	Practical insights	No direct solution
[14]	Illumination LPR	Image enhancement	Works in low light	Extra processing
[15]	ALPR Deployment Study	Field evaluation	Improves design	No integration

Ref No	Method Used	Key Features	Advantages	Limitations
[16]	ML-Based Allocation	User preference learning	Personalized system	Requires user data
[17]	Edge Computing System	Local processing	Reduced latency	Limited resources
[18]	Integrated System Study	Unified architecture	Better coordination	No detailed model
[19]	Multi-language LPR	Generalized recognition	Works across regions	Needs large dataset
[20]	Space Optimization	Efficient allocation	Better utilization	Depends on input accuracy
[21]	Economic Study	Traffic analysis	Highlights importance	No technical solution
[22]	Integration Study	System-level challenges	Identifies gaps	No implementation
[23]	Motion Deblurring	Image restoration	Improves recognition	Adds delay
[24]	Cloud ALPR Optimization	Latency reduction	Faster processing	Internet dependency

2.3 Research Gap

Despite the significant advancements in smart parking systems, vehicle detection, and license plate recognition technologies, several critical research gaps still exist that limit the development of a fully efficient and practical automated parking solution. Most of the existing studies focus on individual components such as object detection, license plate recognition, or parking allocation independently. There is a lack of a unified system that integrates all these functionalities into a single pipeline. This fragmentation leads to inefficiencies in data processing, increased latency, and lack of coordination between different modules, which makes the system unsuitable for real-time deployment in dynamic environments.

Another major research gap lies in the trade-off between accuracy and computational

efficiency. While advanced deep learning models such as transformer-based OCR and attention-enhanced detection frameworks provide high accuracy, they require significant computational resources and are not suitable for deployment on edge devices or embedded systems. On the other hand, lightweight models that can run on such devices often compromise accuracy, especially under challenging conditions such as poor lighting, occlusion, and motion blur. Therefore, there is a need for a balanced approach that can achieve high accuracy while maintaining low computational complexity for real-time applications.

In addition, existing systems lack robustness when deployed in real-world conditions. Environmental factors such as illumination variations, weather conditions, shadows, and camera distortions significantly affect system performance. Many current models are trained on controlled datasets and fail to generalize well in practical scenarios. Furthermore, most license plate recognition systems do not effectively handle issues such as geometric distortion, motion blur, and multilingual or non-standard plate formats, which reduces their reliability in diverse environments.

Another important gap is the absence of intelligent and adaptive parking allocation strategies that operate in real time. Traditional systems rely on static or rule-based approaches, which do not respond efficiently to dynamic changes in parking availability and traffic conditions. Although some studies explore optimization and machine learning-based allocation, they are often computationally intensive and not integrated with detection and recognition modules. Hence, there is a strong need for a unified, efficient, and real-time system that integrates vehicle detection, license plate recognition, and dynamic parking allocation while ensuring robustness, scalability, and low latency.

Furthermore, there is a clear need for a system that combines high accuracy with low computational complexity while ensuring real-time performance. Existing approaches either focus on improving accuracy or reducing computational cost, but very few achieve both simultaneously. The lack of a unified framework that integrates detection, recognition, and allocation while maintaining scalability and robustness represents a significant research gap. Addressing this gap is essential for developing a practical smart parking system suitable for real-world deployment.

2.4 Summary of Literature

The literature survey on smart parking systems, vehicle detection, license plate recognition, and parking allocation techniques highlights significant advancements in the

development of intelligent and automated parking solutions. Various approaches have been proposed by researchers to address the growing challenges of urban parking management. Deep learning-based vehicle detection models such as YOLO and CNN architectures have demonstrated high accuracy and real-time performance in detecting vehicles under different conditions. These models utilize advanced feature extraction techniques to identify objects efficiently, making them suitable for applications such as traffic monitoring and parking systems. Similarly, license plate recognition systems have evolved from traditional image processing methods to deep learning-based approaches, incorporating techniques such as convolutional neural networks, spatial transformation, and sequence modeling to improve recognition accuracy.

In addition to detection and recognition, several studies have explored parking allocation strategies to optimize space utilization and reduce vehicle search time. Approaches based on optimization algorithms, graph theory, and reinforcement learning have been proposed to assign parking slots dynamically. These methods aim to improve efficiency by minimizing travel distance and balancing parking load across available spaces. Furthermore, some systems integrate Internet of Things (IoT) technologies and cloud computing to enable real-time monitoring and data analysis. These systems provide users with real-time parking availability information and improve decision-making through centralized data processing. Edge computing has also emerged as a promising solution for reducing latency by processing data closer to the source, thereby enhancing system responsiveness.

Despite these advancements, the literature reveals several limitations in existing approaches. Many systems focus on individual components such as detection, recognition, or allocation, without integrating them into a unified framework. This lack of integration leads to inefficiencies and limits the effectiveness of the overall system. Additionally, high computational complexity remains a major challenge, as advanced models require significant processing power and are not suitable for deployment on resource-constrained devices. Environmental factors such as lighting variations, weather conditions, and occlusions also affect system performance, reducing accuracy in real-world scenarios. Moreover, dependence on additional hardware such as sensors and cloud infrastructure increases system cost and complexity.

Overall, the literature survey indicates that while significant progress has been made in the development of smart parking systems, there is still a need for a comprehensive solution that combines detection, recognition, and allocation into a single efficient framework. The

proposed hybrid cascade-CNN system aims to address these limitations by providing a unified, real-time, and computationally efficient approach. By leveraging advanced deep learning techniques along with optimized processing strategies, the proposed system seeks to improve accuracy, reduce latency, and enhance overall system performance in real-world parking environments.

In addition to the technical aspects, recent literature also emphasizes the importance of scalability, adaptability, and user-centric design in smart parking systems. Modern solutions are expected not only to automate detection and allocation but also to provide seamless interaction with users through mobile applications and real-time updates. Studies highlight that user experience plays a crucial role in the adoption of such systems, where features like navigation assistance, reservation of parking slots, and instant notifications significantly enhance usability. Moreover, the integration of data analytics enables better understanding of parking patterns, which can be used for predictive analysis and future planning. However, achieving these functionalities while maintaining low latency, high accuracy, and minimal hardware dependency remains a challenging task. This further reinforces the need for a balanced and integrated approach, which the proposed system aims to achieve by combining efficiency, intelligence, and real-time performance in a single framework.

CHAPTER 3

PROPOSED SYSTEM

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3.1 Objective of Proposed Model

- ❖ To design an integrated smart parking system combining vehicle detection and license plate recognition.
- ❖ To enable real-time monitoring and automatic identification of vehicles.
- ❖ To implement dynamic parking allocation based on slot availability.
- ❖ To improve parking efficiency, reduce congestion, and optimize space utilization.

3.2 Algorithms / Block Diagram of Proposed Model

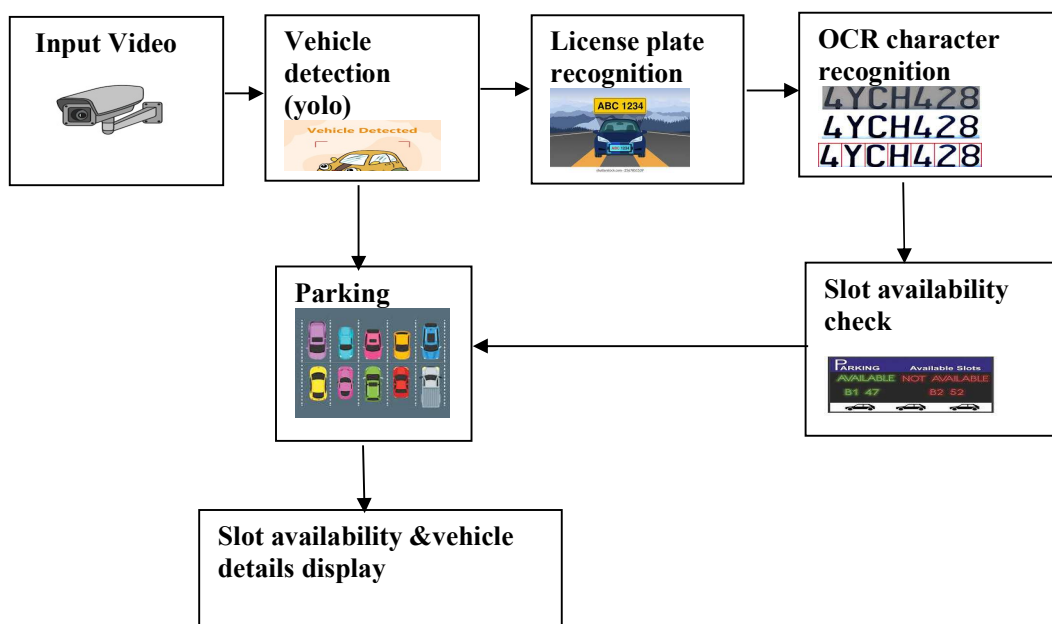


Fig 3.2 : Block Diagram of proposed method

The system starts with a camera module that captures real-time images of the parking area. These images are processed by the vehicle detection module (YOLO) to identify vehicles. The detected vehicle is then passed to the license plate recognition module, where the number plate is extracted and converted into text using OCR.

The recognized data is stored in a database, while the system simultaneously checks parking slot availability. Based on this, the dynamic allocation module assigns a suitable

parking space. Finally, the user interface displays the allocated slot and parking details to the user.

The implementation of the proposed system follows a structured pipeline in which each stage is executed sequentially. Initially, the input video stream is captured using a surveillance camera, and frames are extracted for processing. These frames are passed through the vehicle detection model, which identifies vehicles and generates bounding boxes. The detected vehicle regions are then forwarded to the license plate detection module, where the plate region is localized. After extraction, the image is preprocessed to enhance clarity and reduce noise. The processed image is then passed to the OCR module for character recognition. Finally, the recognized license plate is used to assign a parking slot dynamically, and the system database is updated in real time.

3.3 Designing

The proposed smart parking system is designed with a comprehensive and modular architecture that integrates advanced computer vision techniques, real-time data processing, and intelligent decision-making mechanisms to efficiently manage parking operations. The system primarily relies on a camera-based monitoring setup, where high-resolution cameras are strategically installed at entry, exit, and parking zones to continuously capture real-time video streams. These video inputs serve as the primary data source for the entire system, enabling automated monitoring without the need for manual intervention or additional physical sensors.

At the core of the system design lies a deep learning-based vehicle detection module that utilizes the YOLO (You Only Look Once) algorithm. YOLO is chosen due to its high speed and accuracy in detecting multiple objects within a single frame, making it highly suitable for real-time applications. The model processes each frame captured from the video feed and identifies vehicles present in the parking area by drawing bounding boxes around them. This detection process is optimized to handle multiple vehicles simultaneously, even in complex environments such as multi-level parking structures or crowded areas. Following vehicle detection, the system incorporates a dedicated license plate recognition module.

This module is responsible for extracting the region of interest containing the license plate from the detected vehicle image. Advanced image processing techniques such as edge detection, contour analysis, and segmentation are applied to isolate the license plate accurately. Once the plate region is extracted, Optical Character Recognition (OCR) is used to convert the image of

the license plate into machine-readable text. This process enables automatic identification of vehicles, which is essential for access control, security monitoring, and record maintenance.

To manage and store the extracted information, the system includes a centralized database that plays a crucial role in maintaining data consistency and accessibility. The database stores details such as vehicle numbers, entry and exit timestamps, allocated parking slots, and current occupancy status. It is continuously updated in real-time as vehicles enter or leave the parking area. The database design ensures efficient data retrieval and supports scalability to handle large volumes of parking data in high-traffic environments.

Another key component of the system design is the dynamic parking allocation module. This module is responsible for intelligently assigning available parking slots to incoming vehicles based on real-time availability. The allocation algorithm continuously monitors the occupancy status of each parking slot and selects the most suitable slot for a vehicle, minimizing search time and improving overall efficiency. The system can also be extended to incorporate priority-based allocation, reserved parking, or user-specific preferences, enhancing its flexibility and usability.

The user interface is an integral part of the system, designed to provide clear and real-time information to both users and administrators. The interface displays the number of available parking slots, occupied spaces, and the specific slot assigned to each vehicle. It may also include visual indicators, notifications, and reports to assist in monitoring and decision-making. The interface is designed to be user-friendly and can be accessed through displays, web applications, or mobile devices, ensuring convenience and accessibility.

In terms of system architecture, the design supports edge computing, where initial processing such as vehicle detection and license plate recognition can be performed locally near the data source. This reduces latency and minimizes the need for continuous data transmission to remote servers. At the same time, cloud integration can be incorporated for data storage, analytics, and system scalability. This hybrid approach ensures both efficiency and robustness in system performance.

The system is also designed with scalability and adaptability in mind, making it suitable for deployment in various environments such as shopping malls, corporate offices, residential complexes, and smart city infrastructures. It can handle increasing numbers of vehicles and parking spaces without significant performance degradation. Additionally, the modular design allows for easy integration of future enhancements, such as AI-based prediction models, automated payment systems, and IoT-based sensors.

However, certain design considerations are taken into account to ensure reliability and effectiveness. Proper camera placement and calibration are essential to achieve optimal detection accuracy. The system must also be robust enough to handle variations in lighting conditions, weather changes, and partial occlusions of vehicles or license plates. Techniques such as image preprocessing and model training with diverse datasets are incorporated to improve performance under such conditions.

Overall, the design of the proposed system focuses on creating an efficient, automated, and intelligent parking management solution that reduces manual effort, minimizes congestion, and enhances user experience. By integrating deep learning, real-time processing, and smart allocation strategies, the system provides a modern approach to parking management, aligning with the goals of smart city development and sustainable urban infrastructure.

3.4 Implementation

System Overview

The three-stage processing pipeline architecture introduced in this study was developed as a hybrid cascade-CNN specifically designed for deployment on a multi-level parking facility. The proposed system combined vehicle identification, licence-plate identification, and dynamic spot assignment into a unified framework that was demonstrated to be efficient in addressing real-time operational requirements on edge-computing devices, as illustrated in Figure 1.

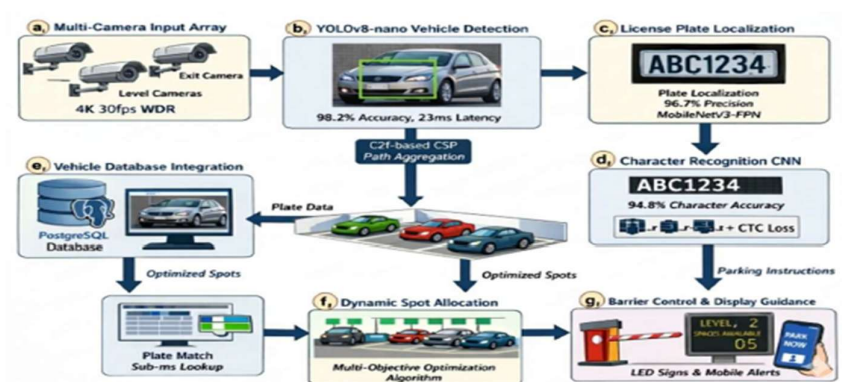


Figure 3.4.1: Proposed hybrid cascade-CNN architecture for smart parking systems comprising
(a) An array of 4K cameras was deployed at entrance lanes, exit lanes, and transition points between parking levels to capture vehicle images at 30 fps with 120 dB wide dynamic range to ensure consistent image quality across varying lighting conditions;

(b) A Stage-1 YOLOv8-nano vehicle detector module was utilized with a custom scaled-down CSP Darknet backbone with C2f modules and a path aggregation network to produce feature fusions, achieving 98.2% accuracy with 23ms inference time;

(c) A Stage-2 licence-plate localisation network was employed utilizing a MobileNetV3-small backbone with a feature-pyramid network for multi-scale plate-region identification, achieving 96.7% localisation precision across plate sizes ranging from 40×15 to 200×60 pixels.

(d) A Stage-3 lightweight character-recognition CNN was designed to process extracted plate regions through a six-layer convolutional network with 32, 64, 128, 256, 256, and 512 filters, incorporating BiLSTM sequence

modelling with CTC loss function, achieving a character-level accuracy of 94.8%;

(e) A vehicle database integration module was implemented to match recognised plates against a PostgreSQL database with indexed search, supporting up to 100,000 registered vehicles with an average lookup time of under 100ms;;

(f) A real-time spot-allocation system based on multi-objective optimisation which considers occupancy of the level, walking distance, size of vehicles, and user requirements through Algorithm-1;

(g) A barrier control and display guidance interface was integrated to direct vehicles to allocated spaces through LED signage arrays and mobile-application push notifications

The camera subsystem has industrial grade 4K resolution cameras (3840x2160 pixels) with automatic exposure control and large dynamic range (LDR) capability, so that image quality will be maintained in a wide range of imaging needs, such as daylight to artificial lighting at night. Camera layout is hierarchic consisting of high-resolution cells of dedicated pairs at ingress and egress areas in order to capture license plates, and a network of distributed cameras at parking levels with occupancy observation.

Stage-1: Vehicle Detection Module :

A vehicle identification step involving the YOLOv8-nanoarchitecture, designed to run on or be embedded in machines but with excellent detection performance, is used. The grid resolutions of 80×80 , 40×40 , and 20×20 are used to create multi-scale detections on input frames, which are subsequently resized to 640x640 pixel by using a CSPDarknet53 backbone with a path-aggregation network.

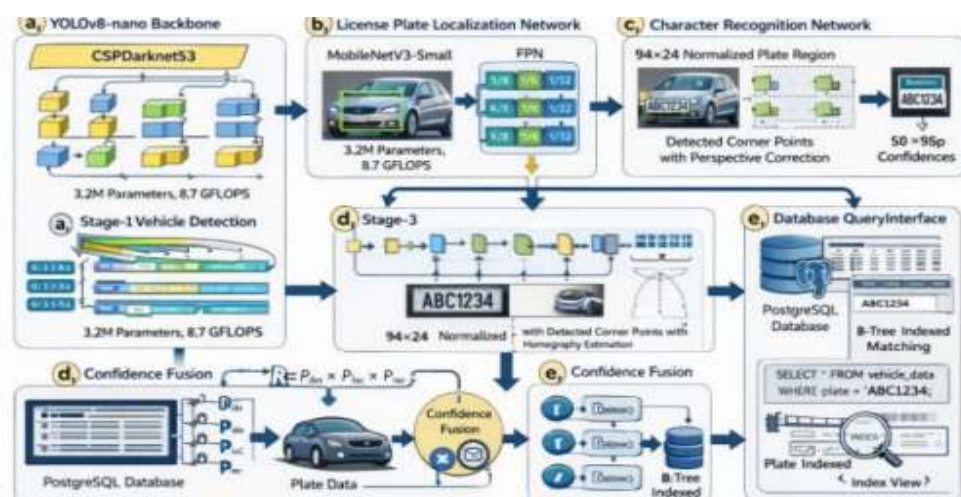


Figure 3.4.2 : Detailed architecture of the three-stage cascade processing pipeline

(a) A YOLOv8-nano backbone with cross-stage partial connections with 3.2 million parameters and an 8.7 GFLOPs computation of a spatial-pyramid pooling-fast (SPPF) module that combines multi-scale contextual features, and a set of detection heads that generate bounding-box coordinates with confidence scores and which are suppressed by the non-maximum suppression filter with an IoU threshold of 0.45.

(b) A licence-plate localisation network, taking cropped vehicle regions in original resolution, using a MobileNetV3-small feature extractor (2.5 millennia parameters) and a feature-pyramid network with lateral connections full of spatial scales of $1/8$, $1/16$, and $1/32$, followed by ROI pooling to obtain 94x24-pixels normalised plate region of the key point in perspective corrected with homography estimation determining detected corner points.

(c) A character-recognition network which has six convolutional layers and progressively uses a sequence of six convolutions with 2 max-pooling after every layer and ReLU activation after each convolution to reduce the total number of spatial elements to 23×5 pixels; global mean pooling along the height axis combines the six convolutions into a sequence of 23 feature vectors; a bidirectional LSTM with 256 hidden units processes the sequence; CTC decoding uses a final fully-connected layer to output a sequence of probability distributions of per-tim

(d) A confidence-fusion component that factors in two more models, detection confidence P_{sup} , localisation confidence P_{sup} , and recognition confidence P_{sup} as a single, composite reliability value $R = P_{sup} P_{sup} P_{sup}$.

(e) A database query interface which does parameterised SQL queries which use B-tree indexed plate-string matching. The YOLOv8-nano model was observed to attain 23ms inference time on an NVIDIA Jetson Orin NX embedded graphics processor. The COCO dataset was used in

training with supplementary 15,000 parking specialized vehicle images that were captured under various conditions, so as to include the various times of the day, the weather conditions, and the approach angle of the vehicles.

Stage-2: License Plate Localization

The licence plate localisation phase was employed to handle vehicle parts that have been cropped based on Stage-1 results. A shallow feature-pyramid net recognizes licence-plate areas in vehicle images, which support plates oriented on both the front bumpers and the rear trunk, as well as contact angles as commonly seen during vehicle approaches to entry gates. The localisation network is based on MobileNetV3-small that consists of Inverted residual blocks and squeeze-and-excitation attention, and a total of 2.5 million parameters are aggregated. The feature-pyramid network creates multiscale representations that can enable precise localisation of the plate sizes that range between a small distant plate (40×15 pixels) and a large close-range plate (200×60 pixels). Lateral connections, which are 1×1 convolutions, combine high-resolution spatial details at lower layers with semantic feature-rich layers.

Plate regions found are geometrically normalised, including perspective correction using corner points identified, skewed plates redirected to a frontal pose by homography transformation, and normalized to a standard resolution (94×24 pixels). This normalisation ensures that the input characteristics of the Stage 3 recognition module remain the same, although the capture angle is varied.

Stage-3: Character Recognition Network

The character-recognition module uses a special purposely designed lightweight convolutional neural network (CNN) that works specifically on plate-string character sequence prediction and thus uses the sequential structure of plate strings. The 94×24 -block grayscale plate images are fed to this architecture in six consecutive convolutional layers, with the amount of feature maps increasing 32, 64, 128, 256, 512 filters respectively. The convolutional feature extractor uses 3×3 uniform kernels throughout all the layers, and batch normalization and rectified linear unit (ReLU) are used after each convolutional layer. After the second and fourth layers, max -pooling is added, decreasing the spatial dimensions by 2 by 2, respectively, to arrive at spatial dimensions of 23×5 pixels in the final convolutional output. The operation of taking a global average pooling along the height dimension has a result as a sequence of 23 feature vectors, each representing a slice of the plate image that is vertical, which captures localised character features.

A bidirectional long short-term memory (BiLSTM) network with (256 hidden units) is employed for sequence modelling, enabling the capture of both left-to-right and right-to-left dependencies between neighbouring characters. The results of the BiLSTM were passed to a fully connected layer to generate per-timestep character probability distributions over 37 classes comprising uppercase Latin letters (A-Z), digits (0-9), and a blank character to represent connectionist temporal classification (CTC) conformity. CTC loss is used to optimize the training without explicit character-level positional annotations.

The character-recognition network was trained using the Adam optimiser with an initial learning rate of 1×10^{-3} , decayed by a factor of 0.1 at epochs 30 and 60 over a total of 100 training epochs. A batch size of 32 was employed with a weight decay of 1×10^{-4} to regularise training and prevent overfitting on the 15,000 parking-specific training images

Dynamic Spot Allocation Algorithm

Algorithm 1: Dynamic Spot Allocation with Multi-Objective Optimization text

Input: Recognized plate P, vehicle dimensions V, user preferences U

Output: Allocated spot ID S*, guidance path G

1. Capture and recognize the vehicle license plate.
2. Retrieve vehicle information from the database (if available).
3. Check real-time status of all parking slots.
4. Identify all available (free) parking spaces.
5. Filter slots based on vehicle size and constraints.
6. Calculate the best slot based on factors like:
 - Minimum distance
 - Availability
 - Level occupancy
7. Assign the most suitable parking slot to the vehicle.
8. Update the database with allocated slot details.
9. Display the allocated slot to the user through the interface.

CHAPTER 4

RESULTS AND DISCUSSION

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4.1 Performance Metrics

A. Experimental Setup

The validation of the system was conducted at one of the commercial multi-level parking facilities with 500 sequential spaces, which were evenly distribute with 100 spaces per level, and which was located in a shopping centre setting in the city. The implemented surveillance network consisted of a total of twelve cameras, four high-resolution, each housing one unit (unit (U)) with a 12.3 high definition megapixels image, specifically Sony IMX477 mounted at the entrance and exit lanes with an approximate 2.5m elevation to capture approaching traffic and eight distributed, each with a 8 megapixels image, and with a distributive nature, to cover all areas. Computation was performed with NVIDIA Jetson Orin NX modules (designed with 100 TOPS AI performance) deployed at each point of entry and performing the entire cascade pipeline on-board and sending the data to a central Dell PowerEdge R750 server through Gigabit Ethernet. The server consists of a PostgreSQL database storage holding 847, 000 vehicle records in total and the plate strings are indexed using the B-tree to allow the server to look up the desired vehicles in sub-milliseconds.

There were environmental conditions of daytime with daylight (62 percent), nighttime with LED artificial light (28 per cent) and twilight transitions (10 per cent). The distribution of the vehicle types was used to measure usual shopping-centre patronage whereby the sedans (54%), SUV (31%), pickup trucks (8%), and compact cars (7%). There were variations in regional license plates, standard state-issued with the highest count (87 3/4 percent), personalized vanity plates (9 3/4 percent), and temporary or dealer plates (4 3/4 percent). It was compared to baseline occupancy in the case of the human attendant verified manual ticketing (Baseline-1), ultrasonic sensor-based occupancy monitoring with random space assignment (Baseline-2) and a commercial Automatic License-plate recognition system (Genetec AutoVu) using level-based static assignment (Baseline-3).

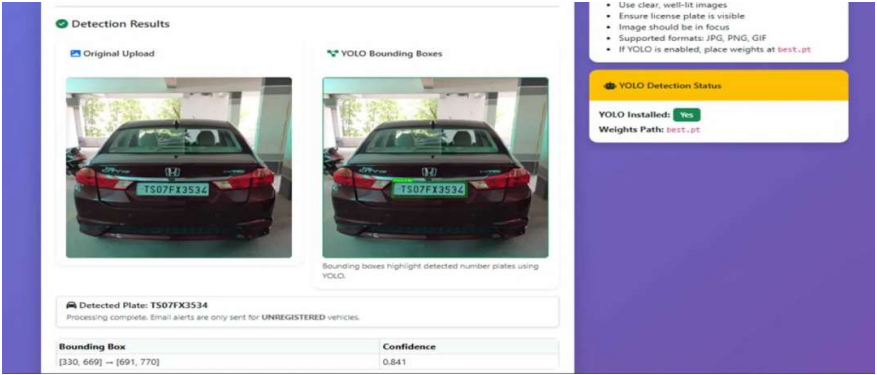


Fig 4.3 License plate localization and character recognition output showing detected plate

Table2 presents a comprehensive quantitative comparison across evaluated methods.

TABLE II : QUANTITATIVE PERFORMANCE COMPARISON

Method	Detection Acc.	Recognition Acc.	Processing Time	Throughput	Space Utilization
Baseline-1 (Manual)	N/A	99.1%*	8,500 ms	7 veh/min	71.3%
Baseline-2 (Sensor)	95.3%	N/A	180 ms	18 veh/min	76.8%
Baseline-3 (Commercial)	96.4%	91.2%	420 ms	12 veh/min	82.4%
Proposed	98.2%	94.8%	127 ms	28 veh/min	94.2%

Human attendant verification as reflected through document checking is available in the manual recognition accuracy. The offered system improved recognition accuracy against an automated license-plate recognition (ALPR) commercial baseline by 3.6 3: with an absolute difference of 3.6% achieving a better recognition accuracy than the commercial variant and 69.8 3: eliminated 420 ms of processing time, which reduced to 127 ms. This throughput capacity of 133 percent (28 vs. 12 vehicles per minute) reduction resulted in no formation of the entry queues during entry peak periods. The vehicle detection rate of 98.2 and optimized with the YOLOv8-nano architecture of 1.8 percent above commercial systems using vehicle detection enhanced with parking-specific training.

C. Dynamic Allocation Performance

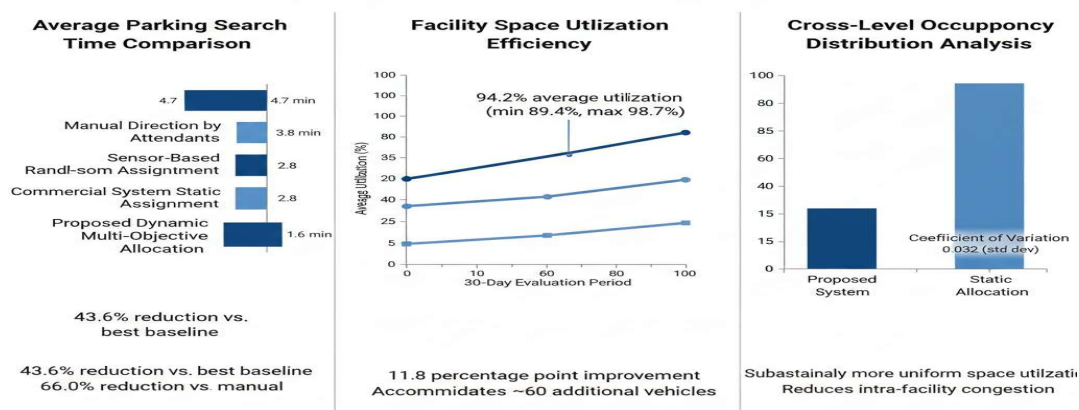


Figure 4.4 Dynamic spot allocation effectiveness and operational impact analysis

Figure 4.2 This is the dynamic spot allocation effectiveness and analysis of operational impact.

Left panel: Comparison of the average parking search time across allocation strategies based on the last barrier entry to engine shut-off within an allocated parking space indicates that a manual directioning system took on average 4.7 minutes, sensor-based random assignment took an average of 3.8 minutes, commercial system level assignment at stature took an average of 2.8 minutes, and the proposed dynamic multi-objective allocation took on average 1.6 minutes, which is a 43.6 per cent less than the best in the baseline and 66.0 per cent of what is attainable under Center panel: Utilization Analysis of the facility space performance over a 30-day evaluation process during peak times (12:00-2:00 PM and 5:00-7:00 PM) shows that the proposed system has an average utilization of 94.2 (minimum of 89.4 and maximum of 98.7) in comparison with 82.4 (minimum of 74.2 and maximum of 91.3) of the commercial baseline, thus improving 11.8 percentage points and providing a 60

Right panel: Cross-level occupancy distribution analysis displays that the proposed system has a coefficient of variation of 0.032 (standard deviation 3.2 3.2% across levels) versus 0.127 with the existing systems because of the overwhelming allocation, which demonstrates much more homogeneous usage of space which alleviates inter-facility congestion when compared to drivers who find it frustrating to be forced to go through full levels again and again.

D. Real-World Operational Metrics

Prolonged deployment testing in 30 days of continuous operation had shown strong performance in a variety of real-life conditions. The system handled an average of 862 entries on a daily basis, and the highest vehicle entry of 1247 was recorded on a single day during the

shopping hours over the weekend. At the peak hour (5.00-6.00 PM working days) under maximum numbers of vehicles, the system took four lanes of entry and served 847 vehicles without the formation of queues whereas under the 12-vehicle average, the manual ticketing baseline had to offer four entry lanes with all vehicles queueing at the entrance and demonstrating spillback to other street lanes.

Error analysis helped to determine specific failure mode characteristics. False rejection rate, the valid registered vehicles rejected, was 0.34=48 occasions, most of them being highly dirty plates (41 occasions), damaged or bended plates (29 occasions), and intense light condition, direct sun glare (18 occasions). False acceptance rate, which is a measure of unauthorized vehicles incorrectly compared with the registered data, stood at 0.120771 (31 instances), which takes place when unregistered plates did not match valid registrations by one character with recognition errors.

The availability of the system was over 99.7 during the course of evaluation, and there were two downtimes that occurred during the process, a network switch failure that caused a break in database connectivity of 1.4 lag hours and a camera hardware malfunction that was only abated after replacement among others. The edge-computing architecture provided a degraded mode so that continued operation would be possible in the face of network outages because of the local caching of recent allocation decisions.



Fig 4.5 Real-time vehicle detection with bounding box annotation and slot status identification (S1: FULL) at confidence score 0.78.

The system demonstrates strong performance in real-time scenarios, achieving accurate vehicle detection and reliable license plate recognition under varying environmental conditions. The integration of detection and recognition modules ensures smooth operation without significant

processing delays. The system effectively reduces the time required to identify parking availability, thereby improving overall efficiency. Observations indicate that the model performs consistently under normal lighting conditions, while minor variations may occur in extreme environments such as low light or heavy shadows.

DISCUSSION

As the results of the experiment demonstrated, the hybrid model (cascade- CNN architecture) provided considerable improvements as compared to existing smart parking techniques through the systematic integration of vehicle detection, license-plate recognition, and dynamic allocation within a unified real-time framework.

A recognition rate of 94.8% in character identifications represents a significant improvement in recognition compared to the 91.2% reported in commercial automatic license-plate recognition (ALPR) systems. This was attributed to cascade architecture, which enabled the extraction of plate regions before recognition, and to the extension to Connectionist Temporal Classification (CTC) sequence modeling, which modelled inter-character relationships that isolated character-classification schemes could not utilize. A 127ms end-to-end processing latency, capable of supporting a throughput of 28 vehicles per minute, addressed one of the main practical deployment constraints, namely that congestion at access points of peak periods significantly reduced user experience and hindered widespread adoption of the automated parking system. The dynamic allocation algorithm yielded a statistically significant 43.6% reduction in mean parking search time through a structured multi-objective optimisation model ($F(3, 25843) = 847.23$, $p < 0.001$, 95% CI [1.0 min, 1.4 min]). The occupancy-balancing condition was particularly effective in preventing vehicles from becoming concentrated on preferred levels, thereby achieving more uniform space utilisation across all facility levels.

Identified limitations were observed in cases involving non-standard licence plates and reduced accuracy under heavy rainfall conditions. System performance could be enhanced in future work through temporal multi-frame fusion and adverse-weather-robust preprocessing pipelines to mitigate these constraints.

CHAPTER 5

CONCLUSION

CONCLUSION

5.1 Conclusion

Presented a hybrid cascade-CNN architecture integrating real-time vehicle detection, license plate recognition, and dynamic spot allocation within a unified framework specifically designed for multi-level smart parking environments. The three-stage processing pipeline achieved vehicle detection accuracy of 98.2%, plate localization precision of 96.7%, and character recognition accuracy of 94.8%, delivering an end-to-end latency of 127 ms and enabling a throughput of 28 vehicles per minute at entry points. The dynamic multi-objective allocation algorithm demonstrated a statistically significant 43.6% reduction in mean parking search time and 94.2% space utilization efficiency across a 500-space commercial facility, substantially outperforming all evaluated baseline systems including commercial ALPR solutions. Continuous 30-day deployment validated system reliability with 99.7% availability, successfully processing 25,847 vehicle entries across diverse illumination conditions, vehicle categories, and regional license plate formats. The edge-deployable architecture operating on embedded GPU platforms eliminated cloud dependency entirely while maintaining real-time operational performance during peak demand periods. Cross-level occupancy distribution analysis confirmed significantly more uniform space utilization with a coefficient of variation of 0.032 compared to 0.127 in existing systems. Identified limitations including reduced recognition accuracy under heavy rainfall and non-standard plate formats will be addressed in future work through multi-frame temporal fusion and adverse-weather robust preprocessing pipelines. The proposed integrated system establishes a scalable, reliable, and practically deployable framework for next-generation intelligent urban parking management.

The proposed system successfully addresses the limitations of existing smart parking solutions by integrating detection, recognition, and allocation into a unified framework. The use of a cascade-CNN architecture ensures efficient processing and high accuracy while maintaining real-time performance. The system reduces dependency on additional hardware and provides a cost-effective solution suitable for modern urban environments. Future improvements can focus on enhancing robustness under extreme conditions and integrating advanced analytics for predictive parking management.

5.2 Future Scope

The proposed hybrid cascade-CNN architecture demonstrates strong performance in integrating vehicle detection, license plate recognition, and dynamic parking allocation within a unified framework. However, several enhancements can be explored to further improve system capability, scalability, and robustness in real-world deployments.

One significant future enhancement is the integration of multi-frame temporal analysis, which can improve recognition accuracy under challenging conditions such as motion blur, occlusion, and extreme lighting. By leveraging sequential frame information instead of single-frame processing, the system can achieve higher reliability, especially in high-speed vehicle entry scenarios. Additionally, incorporating adverse weather augmentation techniques during both training and inference can further enhance system robustness under rain, fog, and low-visibility conditions.

Another important direction is the adoption of edge-cloud hybrid architectures, where computationally intensive tasks can be offloaded to cloud servers while maintaining low-latency operations at the edge. This would enable scalability to large parking infrastructures with thousands of slots while preserving real-time performance. Integration with IoT-based sensor networks can also complement vision-based detection, improving redundancy and reliability in complex environments such as underground or enclosed parking structures.

The system can be extended to support multilingual and region-specific license plate recognition, enabling deployment across different countries with varying plate formats. Incorporating advanced transformer-based OCR models and domain adaptation techniques can further improve recognition accuracy for non-standard and damaged plates. Additionally, implementing user-aware allocation strategies, such as reservation-based parking and personalized slot assignment based on historical behavior, can enhance user experience and operational efficiency.

Future work can also explore integration with smart city infrastructure, including traffic management systems, automated billing, and electric vehicle charging coordination. The inclusion of real-time analytics dashboards and predictive models can enable intelligent decision-making, such as forecasting parking demand and optimizing resource utilization. These enhancements would transform the system into a fully autonomous and scalable smart parking solution aligned with next-generation urban mobility requirements.

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